

**“Production capacity in India is currently estimated in excess of 400 MW for cells and is expected to cross 750 MW by the end of year 2010.—Sathya Prasad, president, SEMI India**



i.e., one house can have one system, having two CFLs of 9W each.

Going forward, the domestic demand is expected to increase dramatically, driven largely by the Jawahar Nehru National Solar Mission. With the policy frameworks of the Ministry of New and Renewable Energy (MNRE), especially the Jawahar Nehru National Solar Mission, and the solar policy and programmes of some states like Gujarat, Haryana and Rajasthan, the grid-connected MW-scale projects market is finally opening up in India.

“Rural electrification and telecom would be the major drivers of growth. Telecom towers for cellphones require a lot of electricity, which is currently provided by diesel generators. This can be replaced with solar power,” says Ajay Prakash Shrivastava, president, Maharishi Solar Technology.

## Technological trends

The first generation of solar cells is based on crystalline silicon (c-Si). The second generation is thin-film solar cells, either Si thin-film or non-Si thin-film (like CdTe and CIS). Dye-sensitised (organic) solar cells (DSSCs) are third-generation. These are seen as a potential candidate as the raw material cost is very less.

“Central Electronics Limited (CEL) has already started work on third-generation DSSCs. For this, it is collaborating with dye cell inventor professor Graetzel of Switzerland-based EPFL. IIT and some national labs are also working towards improving solar cell technologies,” informs Dr V.K. Kaul, general manager, Solar Photovoltaic Group, CEL.

As mentioned before, in India,

## Concentrating solar cells

Researchers in the field of photovoltaics are faced with the challenge of increasing the efficiency of the conversion process while reducing the cost significantly. The solution with the highest cost reduction potential is concentrator photovoltaics (CPV), where the cost is reduced by replacing the expensive PV cell material with lower-cost optical systems covering the receiver aperture. Concentrating solar cells use a lens to concentrate light and increase efficiency.

In recent years, multijunction III-V concentrator solar cells with efficiencies greater than 40 per cent under concentration became available. This situation led to an intense development work on CPV system technology with the result that several of the first market-ready CPV products were installed in power plants in 2008.

The technology has not established a footprint in the Indian market yet.

majority of manufacturers are using crystalline-Si technology. It is a very old and time-tested technology. Thin-film is a good technology but not fully developed for commercial applications and needs more time and R&D to make it better. Thin-film cells are less expensive. The cost of manufacturing them is less as they require less silicon—the

most expensive raw material for solar cells. Also, these don't require direct fall of sunlight. Whether the sun is rising or going down, these still produce electricity. In India, thin-film capacity is available only with Moser Baer.

Thin-film solar cells, however, have some inherent problems. There is a phenomenon called light-induced degradation and efficiency is half that of crystalline Si. So to cater to a particular application, more space is needed for the same power output.

Crystalline-Si is almost fully developed. The only thing needed is to reduce the cost of production through increase in output and efficiency of cells. Right now, the life of a crystalline PV module is more than 25 years.

Thin-film solar cells may be based

on amorphous Si, CIS/CIGS or CdTe material. For amorphous-Si cells, there is scope for making them multijunction. Also, there are prospects for mixing up of materials and putting on different layers. Amorphous-Si results in better efficiency and less degradation. CIGS has almost the same scope. Stability and uniformity of the layer application are the areas of research. CdTe is a poisonous substance, so waste disposal is an issue. After serving the purpose, the modules will have to be recycled.

“Module efficiency of crystalline-Si has gone up from 10 per cent in 1990 to more than 13 per cent today, with the best performers averaging 17 per cent. Cell efficiency has also been on the rise and poly-crystalline cells now have an efficiency of 18 per cent and

**“Domination of crystalline technology is primarily because until now most applications of solar technology in India have been small-scale (traffic lights, streetlights, etc).—Ravi Surapaneni, vice president, Solar Semiconductor**

